# MICROECONOMICS

MICRO-02

**Time:** 5 minutes

**Outcome:** Calculate tax wedges, buyer and seller prices, quantity effects, and revenue.

**Difficulty:** Intermediate

# Tax Wedges and Revenue

## THE CORE IDEA

A per-unit tax creates a wedge between the price buyers pay and the price sellers receive. The wedge reduces the quantity traded because some exchanges whose benefits formerly exceeded their costs no longer occur. Tax revenue equals the per-unit wedge multiplied by the post-tax quantity, while the lost mutually beneficial trades create deadweight loss.

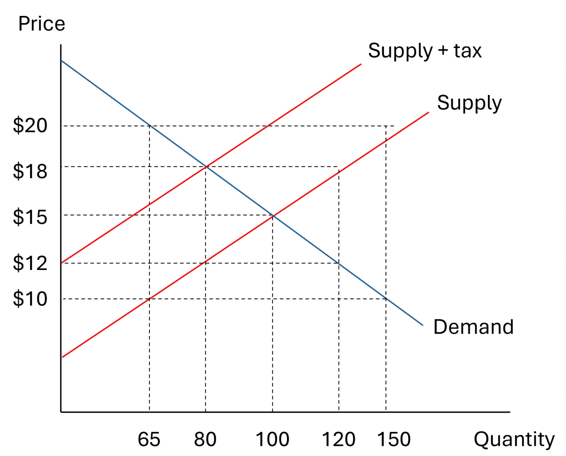
## HOW TO RECOGNIZE IT

- Identify the tax wedge between the price buyers pay and the price sellers receive.
- Use the post-tax quantity, not the original equilibrium quantity, when measuring revenue.
- Tax revenue equals the per-unit tax multiplied by the quantity traded after the tax.
- A tax normally reduces quantity traded and creates deadweight loss from forgone exchanges.

## WATCH OUT

Do not use pre tax quantity or buyer price change only. The tax is $46 - $34 = $12. Tax revenue is $12 × 420 = $5,040.

## WORKED EXAMPLE: CALCULATING TAX REVENUE




The untaxed equilibrium is $15 at 100 units. After the tax, buyers pay $18, sellers receive $12, and quantity falls to 80. The tax wedge is $18 - $12 = $6 per unit, so government revenue is $6 x 80 = $480.

## CHECK YOURSELF

Buyers pay $18 after a tax and sellers receive $12. What is the tax per unit?

**READY?** Return to the game and master this concept.